

CONCURRENCY ARENA - KAPSAMLI OTURUM RAPORU

Tarih: 20.08.2026 | 5 Eszamanlilik Stratejisinin Genel Kiyaslama ve Detayli Kullanici Loglari

Genel Test Parametreleri: 10 Bilet Kapasitesi vs. 20 Eszamanli Kullanici (VU)

Strateji / Model	Satilan/Kota	Veri Tutarlilik	Throughput	Ort. Sure	P95 Gecikme	Toplam Sure
01. Naive (K o r u m a s 1 z)	5 / 10	Kusursuz Tutarlilik	125.79 req/s	102.0 ms	143.0 ms	0.16s
02. Pessimistic Lock	10 / 10	Kusursuz Tutarlilik	104.17 req/s	139.6 ms	176.0 ms	0.19s
03. Optimistic Lock	10 / 10	Kusursuz Tutarlilik	77.52 req/s	169.2 ms	247.0 ms	0.26s
04. Redis Lua Scripting	10 / 10	Kusursuz Tutarlilik	145.99 req/s	95.9 ms	133.0 ms	0.14s
05. Kafka Event-Driven	10 / 10	Kusursuz Tutarlilik	215.05 req/s	64.9 ms	77.0 ms	0.09s

Mimari Sonuclar & Cikarimlar

- Naive: Kilit veya izolasyon olmadigindan Lost Update ve Cifte Satis (Overselling) yasanir.
- Pessimistic: SELECT FOR UPDATE satir kilidi ile %100 tutarlidir, siradaki istekler bekledigi icin latency yuksektir.
- Optimistic: High contention ortaminda JPA @Version cakisir ve 50x retry yukunden dolayi P95 tavan yapar.
- Redis Lua: Stok bellek uzerinde atomik dusurulugunden DB kilitleri kalkar, yuksek Throughput saglanir.
- Kafka: Istekler 202 Accepted ile aninda kuyruğa devredilir, böylece en yuksek Throughput ve en dusuk gecikme elde edilir.

1. 01. Naive (Korumasız) - Varis Sirasi & Kullanici Sırası

Throughput: 125.79 req/s | Avg: 102.0 ms | P95: 143.0 ms | Satılan: 5/10

Kullanici	Istek Zamani	Gecikme	Sonuc	HTTP Kodu	Sunucu Yaniti / Tracking
User #1	16:19:01.154	53 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #2	16:19:01.155	50 ms	BILET ALINDI	HTTP 200	Naive Ticket booked successfully.
User #3	16:19:01.156	82 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #4	16:19:01.156	76 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #5	16:19:01.157	77 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #6	16:19:01.158	69 ms	BILET ALINDI	HTTP 200	Naive Ticket booked successfully.
User #7	16:19:01.159	79 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #8	16:19:01.160	86 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #9	16:19:01.161	94 ms	BILET ALINDI	HTTP 200	Naive Ticket booked successfully.
User #10	16:19:01.162	108 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #11	16:19:01.163	112 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #12	16:19:01.164	106 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #13	16:19:01.165	110 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #14	16:19:01.165	115 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #15	16:19:01.166	120 ms	BILET ALINDI	HTTP 200	Naive Ticket booked successfully.
User #16	16:19:01.167	143 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #17	16:19:01.168	137 ms	BILET ALINDI	HTTP 200	Naive Ticket booked successfully.
User #18	16:19:01.169	141 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #19	16:19:01.170	142 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.
User #20	16:19:01.171	140 ms	KAPASITE DOLU	HTTP 500	An unexpected error occurred on the server.

2. 02. Pessimistic Lock - Varis Sirasi & Kullanici Sonuclari

Throughput: 104.17 req/s | Avg: 139.6 ms | P95: 176.0 ms | Satilan: 10/10

Kullanici	Istek Zamani	Gecikme	Sonuc	HTTP Kodu	Sunucu Yaniti / Tracking
User #1	16:19:02.206	82 ms	Bilet Alindi	HTTP 200	Pessimistic Ticket booked successfully.
User #2	16:19:02.207	71 ms	Bilet Alindi	HTTP 200	Pessimistic Ticket booked successfully.
User #3	16:19:02.208	111 ms	Bilet Alindi	HTTP 200	Pessimistic Ticket booked successfully.
User #4	16:19:02.209	98 ms	Bilet Alindi	HTTP 200	Pessimistic Ticket booked successfully.
User #5	16:19:02.210	121 ms	Bilet Alindi	HTTP 200	Pessimistic Ticket booked successfully.
User #6	16:19:02.210	129 ms	Bilet Alindi	HTTP 200	Pessimistic Ticket booked successfully.
User #7	16:19:02.211	86 ms	Bilet Alindi	HTTP 200	Pessimistic Ticket booked successfully.
User #8	16:19:02.212	137 ms	Bilet Alindi	HTTP 200	Pessimistic Ticket booked successfully.
User #9	16:19:02.213	145 ms	Bilet Alindi	HTTP 200	Pessimistic Ticket booked successfully.
User #10	16:19:02.214	153 ms	Bilet Alindi	HTTP 200	Pessimistic Ticket booked successfully.
User #11	16:19:02.215	155 ms	Kapasite Dolu	HTTP 400	Insufficient seat capacity!
User #12	16:19:02.216	158 ms	Kapasite Dolu	HTTP 400	Insufficient seat capacity!
User #13	16:19:02.217	161 ms	Kapasite Dolu	HTTP 400	Insufficient seat capacity!
User #14	16:19:02.217	162 ms	Kapasite Dolu	HTTP 400	Insufficient seat capacity!
User #15	16:19:02.218	166 ms	Kapasite Dolu	HTTP 400	Insufficient seat capacity!
User #16	16:19:02.219	168 ms	Kapasite Dolu	HTTP 400	Insufficient seat capacity!
User #17	16:19:02.220	169 ms	Kapasite Dolu	HTTP 400	Insufficient seat capacity!
User #18	16:19:02.221	171 ms	Kapasite Dolu	HTTP 400	Insufficient seat capacity!
User #19	16:19:02.221	173 ms	Kapasite Dolu	HTTP 400	Insufficient seat capacity!
User #20	16:19:02.222	176 ms	Kapasite Dolu	HTTP 400	Insufficient seat capacity!

3. 03. Optimistic Lock - Varis Sirasi & Kullanici Sonuclari

Throughput: 77.52 req/s | Avg: 169.2 ms | P95: 247.0 ms | Satilan: 10/10

Kullanici	Istek Zamani	Gecikme	Sonuc	HTTP Kodu	Sunucu Yaniti / Tracking
User #1	16:19:03.304	50 ms	BILET ALINDI	HTTP 200	Optimistic Ticket booked successfully.
User #2	16:19:03.305	144 ms	BILET ALINDI	HTTP 200	Optimistic Ticket booked successfully.
User #3	16:19:03.306	175 ms	BILET ALINDI	HTTP 200	Optimistic Ticket booked successfully.
User #4	16:19:03.307	230 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #5	16:19:03.307	241 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #6	16:19:03.308	186 ms	BILET ALINDI	HTTP 200	Optimistic Ticket booked successfully.
User #7	16:19:03.309	67 ms	BILET ALINDI	HTTP 200	Optimistic Ticket booked successfully.
User #8	16:19:03.310	83 ms	BILET ALINDI	HTTP 200	Optimistic Ticket booked successfully.
User #9	16:19:03.311	95 ms	BILET ALINDI	HTTP 200	Optimistic Ticket booked successfully.
User #10	16:19:03.311	110 ms	BILET ALINDI	HTTP 200	Optimistic Ticket booked successfully.
User #11	16:19:03.312	125 ms	BILET ALINDI	HTTP 200	Optimistic Ticket booked successfully.
User #12	16:19:03.313	221 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #13	16:19:03.314	150 ms	BILET ALINDI	HTTP 200	Optimistic Ticket booked successfully.
User #14	16:19:03.315	247 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #15	16:19:03.316	237 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #16	16:19:03.317	183 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #17	16:19:03.318	195 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #18	16:19:03.319	205 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #19	16:19:03.320	214 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #20	16:19:03.321	225 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!

4. 04. Redis Lua Scripting - Varis Sirasi & Kullanici Sonuclari

Throughput: 145.99 req/s | Avg: 95.9 ms | P95: 133.0 ms | Satilan: 10/10

Kullanici	Istek Zamani	Gecikme	Sonuc	HTTP Kodu	Sunucu Yaniti / Tracking
User #1	16:19:04.465	40 ms	BILET ALINDI	HTTP 200	Redis Ticket booked successfully.
User #2	16:19:04.466	70 ms	BILET ALINDI	HTTP 200	Redis Ticket booked successfully.
User #3	16:19:04.467	97 ms	BILET ALINDI	HTTP 200	Redis Ticket booked successfully.
User #4	16:19:04.467	87 ms	BILET ALINDI	HTTP 200	Redis Ticket booked successfully.
User #5	16:19:04.468	133 ms	BILET ALINDI	HTTP 200	Redis Ticket booked successfully.
User #6	16:19:04.469	106 ms	BILET ALINDI	HTTP 200	Redis Ticket booked successfully.
User #7	16:19:04.469	56 ms	BILET ALINDI	HTTP 200	Redis Ticket booked successfully.
User #8	16:19:04.470	76 ms	BILET ALINDI	HTTP 200	Redis Ticket booked successfully.
User #9	16:19:04.471	123 ms	BILET ALINDI	HTTP 200	Redis Ticket booked successfully.
User #10	16:19:04.472	116 ms	BILET ALINDI	HTTP 200	Redis Ticket booked successfully.
User #11	16:19:04.473	87 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #12	16:19:04.473	92 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #13	16:19:04.474	95 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #14	16:19:04.475	97 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #15	16:19:04.477	102 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #16	16:19:04.478	106 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #17	16:19:04.478	102 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #18	16:19:04.479	112 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #19	16:19:04.480	110 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #20	16:19:04.482	111 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!

5. 05. Kafka Event-Driven - Varis Sirasi & Kullanici Sonuclari

Throughput: 215.05 req/s | Avg: 64.9 ms | P95: 77.0 ms | Satilan: 10/10

Kullanici	Istek Zamani	Gecikme	Sonuc	HTTP Kodu	Sunucu Yaniti / Tracking
User #1	16:19:05.507	56 ms	BILET ALINDI	HTTP 202	Track: 7b1c94d4-9f5...
User #2	16:19:05.508	47 ms	BILET ALINDI	HTTP 202	Track: 5188fd5a-edd...
User #3	16:19:05.509	47 ms	BILET ALINDI	HTTP 202	Track: 7016de9c-13b...
User #4	16:19:05.510	59 ms	BILET ALINDI	HTTP 202	Track: 1314db6e-6e1...
User #5	16:19:05.510	59 ms	BILET ALINDI	HTTP 202	Track: e1b8c484-cd8...
User #6	16:19:05.511	64 ms	BILET ALINDI	HTTP 202	Track: 145a7873-2b2...
User #7	16:19:05.511	55 ms	BILET ALINDI	HTTP 202	Track: b1c4094e-ed9...
User #8	16:19:05.512	55 ms	BILET ALINDI	HTTP 202	Track: b56a9e96-9c5...
User #9	16:19:05.512	64 ms	BILET ALINDI	HTTP 202	Track: 4e535b68-414...
User #10	16:19:05.513	65 ms	BILET ALINDI	HTTP 202	Track: d144f385-00a...
User #11	16:19:05.514	67 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #12	16:19:05.515	66 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #13	16:19:05.516	70 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #14	16:19:05.517	70 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #15	16:19:05.518	73 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #16	16:19:05.519	74 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #17	16:19:05.520	77 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #18	16:19:05.521	76 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #19	16:19:05.522	77 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!
User #20	16:19:05.523	76 ms	KAPASITE DOLU	HTTP 400	Insufficient seat capacity!